Introduction to Articulate Presentations

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Whatever business we're in, whether we're managers, financial analysts, or university professors, we are very likely to suffer through frequent PowerPoint presentations. We see them everywhere—not only at meetings but also online. If we stop to think about it, most of these presentations are dreadfully put together, and either they confuse us or attempt to lull us into accepting weak arguments uncritically. PowerPoint presentations are now parodied, disparaged, and blamed for failures to communicate clearly.

As a professor of psychology who has written widely about how humans process information, some years ago I found myself paying as much attention to the medium as to the message. I began keeping a log of the problems in PowerPoint presentations I saw, and I realized that virtually all of them occur because the presentations failed to respect fundamental characteristics of how we humans perceive, remember, and comprehend information.

I decided to write this book to make the relevant psychological principles clear to people who didn't study these fundamental characteristics of the human mind, and to show how anyone can use these principles to make crystal clear, compelling presentations. This book is fundamentally different from all other books on creating PowerPoint presentations. My recommendations not only are firmly rooted in well-known psychological principles, but I also show you, the reader, how to use these principles to create presentations that communicate clearly and effectively—that are clear and to the point.

When I speak of "psychological principles," I'm not referring to our understanding of neurotic tendencies or any of the things Freud might have discussed during his sessions with patients on the couch. Rather, I am referring to the fruits of the modern science of psychology, which aims in part to discover how perception, memory, comprehension, cognition, motivation, and emotion operate. With the advent of cognitive psychology (which seeks to understand such mental processes by analogy to the operation of computer programs) in the second half of the 20th century and the more recent emergence of cognitive neuroscience (which aims to discover how the brain gives rise to mental processes), we now know more than enough to apply our knowledge to the design of multimedia presentations.

The recommendations offered in this book will help you to make and deliver an effective presentation, step by step—but don't think I'm going to tie you down with rules that will make your presentation formulaic, dry, or lacking in personal style. Your creativity will have plenty of leeway. At the same time, if you design your presentation according to the recommendations I offer here, you will play to the strengths of those who see and hear it, and will avoid being defeated—that is, misunderstood or ignored—because of the inherent weaknesses of human mental processes.

The aims of this book rest on the idea that there's nothing fundamentally wrong with the PowerPoint program as a medium; rather, I claim that the problem lies in how it is used. This view is not universally shared, and you might wonder whether I can deliver on my promises. For example, Edward Tufte, the modern guru of visual communication, claims that the PowerPoint program is inherently flawed. He has found the problems with this tool so pervasive and destructive that he challenges the very idea of using it to communicate.

Before accepting Tufte's verdict, however, consider an analogy: Both an assault rifle and a hammer can be used to kill, but the assault rifle *invites* mayhem, whereas this is an incidental application of a hammer. With regard to its capacity for ruining a presentation, is the PowerPoint program more like the rifle or the hammer? Unlike Tufte, I don't see it as akin to the weapon, which invites being used in the service of destruction. The PowerPoint program is much more like a hammer. Or, more accurately, it's more like a word processing program than an assault rifle. Just as you wouldn't blame Microsoft Word for every bad article you've read, you shouldn't blame the PowerPoint program for every bad presentation you've seen.

In fact, this medium is a remarkably versatile tool that can be extraordinarily effective. Rather than the program's being fundamentally flawed, the problem is that some users, like kids in a candy store, become gluttonous consumers of the options presented by the PowerPoint program—and forget to focus on nutrition. We should use these presentations not to razzle-dazzle with eye candy but to give the meat of our ideas. In this book I provide clear instructions that you can follow to make your ideas crystal clear and immediately comprehensible. But I also go deeper than that—I aim to help you develop intuitions about how to use this powerful tool effectively.

Three Goals and How to Achieve Them

The advice I offer in this book will help you build and deliver presentations that accomplish three goals that virtually define an effective presentation:

Goal 1: *Connect with your audience.*
Your message should connect with the goals and interests of your audience.

Goal 2: *Direct and hold attention.*
You should lead the audience to pay attention to what's important.

Goal 3: *Promote understanding and memory.*
Your presentation should be easy to follow, digest, and remember.

These obviously are worthy goals. The trick is how to achieve them—and that's what this book is about. You can achieve each of these goals if you respect the eight psychological principles that I summarize beneath their applicable goals (and describe in detail in the Appendix). These principles will in turn lead to very specific recommendations in the chapters that follow.

In this introductory chapter, my aim is to give you an overview and a general sense of the eight principles that underlie my advice. We now consider the three goals. When turning to each goal, I introduce the principles that will help to accomplish it.

Goal 1: Connect with Your Audience

You'll communicate effectively only when you focus the audience's attention and interest on a specific message. When I first began to speak publicly, I thought that the goal was to convey as much information on a topic as possible and utterly failed to connect with the audience; I would present many summaries and examples and swamp the audience, not knowing what was of interest to them. I was giving talks to myself, not to the members of the audience (which was doubly unfortunate, since I already knew what I had to say!). You need to speak *to* your audience, not speak *at* them.

Two principles will help you achieve this goal.

Principle 1. The Principle of Relevance

Communication is most effective when neither too much nor too little information is presented.

Abraham Lincoln was once asked how long a man's legs should be. His answer: Long enough to reach the ground. A talk is like a man's legs: It should be as long, but no longer, than necessary to get the job done—in this case, to convey your message. I've never once heard somebody complain that a presentation was too short.

This principle has two main aspects:

1. A presentation must be built from the outset around your take-home message; every aspect of the presentation should be relevant to what you want the audience to know and believe when they walk out the door. We don't want to put ourselves in the position of Yogi Berra when he declared "I didn't really say everything I said."
2. The audience should be told only what they must know to get your message: Telling them too little will leave them puzzled—and telling them too much will leave them overwhelmed, disoriented, and irritated (and ultimately bored, because they will stop trying to track what you are saying). Telling people too much is like what used to happen when digital watches first appeared on the scene: When asked the time, people sometimes—unhelpfully and annoyingly—reported it down to the second. This was neither informative nor humorous.

Example: If you are talking about a *general* difference between Japanese and U.S. consumers, don't break the information down by gender or region.

Principle 2. The Principle of Appropriate Knowledge

Communication requires prior knowledge of pertinent concepts, jargon, and symbols.

You've no doubt often heard that you should "know your audience," and this advice is rock solid: To communicate effectively, your presentation must be pitched at the right level for the audience you wish to reach. The Principle of Relevance is about *what* to communicate, whereas the Principle of Appropriate Knowledge is about *how* to convey that information. Consider these two quotes:

He has never been known to use a word that might send a reader to the dictionary.

—*William Faulkner (about Ernest Hemingway)*

Poor Faulkner. Does he really think big emotions come from big words?

—*Ernest Hemingway (about William Faulkner)*

Hemingway was right in more ways than he may have realized: If a reader is sent to the dictionary, she might just as soon put down the book and walk away. To reach your audience, you need to pitch your talk at a level that makes contact with what the audience already knows. This is true in terms of three factors:

1. The *language* you use—not only the jargon you choose, but also the sophistication and complexity of both the vocabulary and syntactic structure.
2. The types of *displays* you use. For example, a standard bar graph is familiar to (almost) everyone, but a box-and-whisker chart (which typically shows the median, the range of values in the quartile above and below it, and the extreme values) is not universally understood; in fact, you—along with most people—have probably never even heard of it. If you don't recognize the format, you can empathize with how a member of the audience would feel if you violated the Principle of Appropriate Knowledge.
3. The specific *concepts* and information you draw upon. For example, everyone understands amount, but not everyone knows what a first or second derivative is.

If you assume that the members of the audience know more than they actually do and you use unfamiliar language, displays, or concepts, you will not connect with them—and actually might intimidate them. If you assume that they know less than they actually do, you'll

bore them. In either case, you'll fail to connect; even if out of politeness they do manage to keep their eyes propped open and feign not falling asleep.

In short, you have three options: Go over the heads of the audience members and be perceived as a high-falutin' person who likes to hear himself or herself talk; talk down to the audience, and be taken as a patronizing boor; or think about the audience in advance and pitch your talk appropriately. The choice is clear.

In addition, you need to know your audience because your talk should address *their concerns and interests*. It's a curious fact of memory that the more we reflect on material, the more likely it is that the material will stick, that we will remember it, whether or not we've tried to do so. If you can interest your audience and engage them with the material, they will end up remembering it automatically because they will have thought about it.

Example: If you are talking about the economics of oil production to your local Kiwanis group, link the abstract ideas to facets of everyday life, such as the prices of gas, fresh produce, and even various types of synthetic clothing.

Goal 2: Direct and Hold Attention

In the bad old days, before the PowerPoint program existed, speakers often would distribute handouts at the beginning of a presentation. The members of the audience then did exactly what you would expect them to do: They began to read the handout, ignoring what the speaker was saying. Adding insult to injury, they would also rustle the pages as they read, which distracted anyone who was trying to follow the presentation itself.

The modern equivalent of passing out handouts occurs when you give the viewers options about what to pay attention to on a slide. Instead of listening to you, the audience will ignore you while busily examining the slide, reading the text, and looking at the pictures. Because a presentation marches on over time, it is not like a book or paper, where the reader can set his or her own pace and go back when desired. You need to present material in a way that leads the audience through it step by step, and that does not result in their getting lost, confused, or overwhelmed.

In order for your PowerPoint presentation to communicate effectively, it must hold your audience's attention effectively. The audience should be like a boat pushed by an outboard motor, with you supplying the gas! Your talk should *not* leave the audience like so many logs drifting in the current. By using the three psychological principles noted below, in combination with the two just described,

the audience will automatically pay attention to you and your message—simply because that's the path of least resistance.

Principle 3: The Principle of Salience

Attention is drawn to large perceptible differences.

Our brains are "difference detectors." We cannot help being drawn to large differences in perceptible qualities—be they the sudden movement of a rabbit in tall grass, a bright or bold pattern (**like this one**) on a dim background, or loud noises in the kitchen. Salient stimuli are those that are clearly different from the surrounding stimuli, and salient stimuli hijack the audience's attention.

> **Example:** To make the title of a slide the most salient, make it **larger and bolder** than the other text. Similarly, to highlight the most important bar in a graph, color it brighter than the others.

The Principle of Salience is important in part because of what it leads us not to do: Namely, irrelevancies should not be salient.

> **Example:** The PowerPoint program includes many infelicitous background patterns, which at first glance may be tempting eye candy. Some have conspicuous background patterns, such as beach balls and other large, multicolored objects, that grab the viewers' attention. But you don't want the audience to be lost in admiration of the background of your slides. You also don't want the audience to struggle to get your message amid gripping irrelevancies. Instead, the most important part of a slide should command their immediate attention.

This principle also leads us to prioritize different aspects of the message, highlighting what is most important at any given moment in time. Imagine that you are guiding a blindfolded person along a route by giving verbal directions: At the outset, you need to give the person an overall description of the lay of the land ("we'll be going half a mile, with about a dozen turns"), and then only tell her at the relevant times what she needs to know to complete each leg of the journey. Your audience isn't blindfolded, but needs similar guidance.

> **Example:** With the PowerPoint program, you can call attention to text or pictures by sending them in from the side; conversely, you can remove objects from the viewer's attention by making them only a shade or two different from the background color—a process that I will call "graying it out."

Principle 4: The Principle of Discriminability

Two properties must differ by a large enough proportion or they will not be distinguished.

Salience is crucial for directing and holding attention, but a change doesn't even have a chance of being paid attention to if it's not discriminable. Discriminability occurs when we can tell two things apart, whereas salience occurs when differences are so large that attention is involuntarily grabbed by the stand-out object or event. A difference of two inches in height between two toddlers is easy to notice (and hence such a difference is salient), but is essentially invisible (and hence not discriminable) when we compare two pro basketball players who are more than seven feet tall. Our visual systems register relative proportions, not absolute amounts, so that if a proportion is too small, we won't be able to distinguish between two properties. If a change is not discriminable, it may as well not be there.

Example. We've all had the experience of squinting from the back of the room, trying to read a tiny label: The words are too small for us to distinguish the individual letters. Similarly, consider again those fancy and elaborate backgrounds that the PowerPoint program so conveniently provides, which range from cosmic scenes to complex geometric patterns. These backgrounds can make reading text the visual equivalent of sorting wheat from chaff, and can turn your slide into a puzzle to be solved. Even if you assume that they will bother, why make your audience work so hard?

Principle 5: The Principle of Perceptual Organization

People automatically group elements into units, which they then attend to and remember.

You can direct your audience's attention in part by how you arrange the material on your slide; viewers will consider together material that is organized into a single group. We see groups formed from individual elements all of the time. For instance, we see a flock of geese heading south for the winter not as individual birds, but instead as a unit; we see reflectors on the highway not as single pieces of plastic, but instead as a line. The same principles that lead us to organize geese or reflectors are also at work when we hear a presentation and see slides. We automatically group together:

- Objects that are near each other: We see XXX XXX as two groups of 3 Xs, whereas we see XX XX XX—the same number of X marks, differently arranged—as three groups of 2 Xs.
- Objects that appear to be similar: We see XXXOOO as two groups.
- Objects that line up on a smooth curve. We see ——————as a single line, but ——— _ _ _ as two lines (each of which is a group).

- Objects that move similarly: We see two words flying in from the left side of the display as grouped together.
- Objects that form simple shapes. We see [_] as a single group, but not _] [

Example: Labels should be closer to the object (or, in a graph, to the wedge, line, or bar) that they label than to anything else.

Goal 3: Promote Understanding and Memory

If you take advantage of how mental processes operate, you can help the audience easily understand and retain material. Conversely, if you ignore key facts about these processes and their limitations, you will easily confuse or overwhelm the processes. The following three principles will help you promote understanding and retention (when combined with the five summarized earlier).

Principle 6: The Principle of Compatibility

A message is easiest to understand if its form is compatible with its meaning.

Our mental processes tend to make a direct connection between the properties of what we see and hear, and the content of the message being conveyed. In other words, we infer content from form. The admonition not to judge a book by its cover is necessary precisely because we have this tendency.

Example. A vivid illustration of this principle was provided by John Ridley Stroop in 1935. Stroop showed people the names of colors written in various colors of ink. For example, the word "red" was written in red, blue, or green ink; the word "blue" in red, blue, or green ink, and so on. When participants were asked to report the color of the ink (not to read the word), they took more time and made more errors when the word named a color *different* from the color of the ink than when the word named a color the *same* as the color of the ink. Similar interference occurs when people are asked to judge the relative sizes of the words large and small written in small and large typefaces, respectively. We are confused when the two messages—from the form itself and from the meaning—conflict.

A particularly important aspect of this principle is our unconscious and automatic tendency to follow the dictum that *More is More*. This tendency is very clear in graphs, where an additional amount of a wedge or a bar is seen to represent an additional quantity, and a continuous rise and fall of a line is naturally taken to reflect a continuous rise and fall in the entity being measured.

Example. I once saw an egregious violation of the dictum that More is More: The artist presented a bar graph to convey the burglary rates of different cities. Each horizontal bar was made up of little pictures of houses, and longer bars represented larger numbers of houses. So far, so good. But here is the problem: Each bar, regardless of length, showed only one burglarized house (in red)—which meant that the *longer* the bar, the *lower* the rate (because the one burglarized house was a smaller percentage of the whole bar). At first glance, the reader saw the longer bars as indicating more burglaries (which was, after all, the subject of the graph) and had to fight the intuition that *more is more* to realize that longer bars actually meant "fewer." The graph would have been more effective if longer bars had simply indicated a greater percentage of burglaries.

Principle 7: The Principle of Informative Changes

People expect changes in properties to carry information.

This principle has two aspects:

1. When we see or hear a change, we expect it to mean something, so every visible or auditory change should convey information. This idea runs counter to the habits of many PowerPoint users, who include decorations or interesting (but essentially random) visual changes, thinking it makes the talk more attractive. But if words, shapes, or effects don't convey information, they distract.

Example: Telling a joke in the middle of a presentation can give the audience time to "come up for air" (which is a good idea when the presentation is complex). But the joke must be related to the topic of the presentation. For instance, let's say I tell the following joke now (adapted from one found at www.coolfunnyjokes.com/Funny-Jokes/Animal-Jokes/ Vampire-bat.html): A vampire bat, covered in fresh blood, parked himself on the roof of the cave to get some sleep. Pretty soon all the other bats smelled the blood and began hassling him about where he got it. He told them to go away and let him get some sleep, but they persisted until finally he gave in. "OK, follow me" he said and flew out of the cave with hundreds of bats behind him. Down through the valley they swept, across a river and into a forest full of trees. Finally he slowed down and all the other bats excitedly milled around him. "Now, do you see that tree over there?" he asked. "Yes, Yes, Yes!" the bats all screamed in a frenzy. "Good," said the bat, "because I sure as hell didn't!"

Whatever its merits as humor, this joke makes no sense in the current context. If I had presented it in the discussion of discriminability, it could have helped to make the point (sort of). Or I could have used it to illustrate a violation of the Principle of Relevance. But here, it just plain doesn't fit.

This leads to the second aspect of the Principle of Informative Changes:

2. Every change in meaning should be conveyed by a change in appearance. What they see (and hear) is what they get; and what you want them to get should be signaled by something you show (or say).

Example: If part of a graph is a projection to the future, it should look different (perhaps darker or over a colored screen) from the part that summarizes actual data.

Principle 8: The Principle of Capacity Limitations

People have a limited capacity to retain and to process information, and so will not understand a message if too much information must be retained or processed.

We can hold only so much in mind and perform so many mental operations before we reach our limits. You've no doubt had the experience of over-taxing your computer, for example, by trying to download an enormous file. But unlike the tireless, steady machine, we humans are not so patient, and usually will simply give up trying if we must work too hard to decode a message or understand a presentation.

This principle has two parts:

1. We like to joke about our memory foibles (e.g., "A clear conscience is usually the sign of a bad memory"), but the truth is that people can retain only a limited amount as they try to soak in new information. In fact, the audience cannot hold in mind more than four groups of information at once—which I call the *Rule of Four*. But each group itself can contain four groups.

Example: Showing all 64 boxes in a complex organizational chart will overwhelm the audience. But the audience can absorb this information if you build up the slide a portion at a time. For instance, you can present the top layers (where proximity will group them first), and then show successive lower layers on the slide (graying out the upper layers, to direct attention to the new material—but keeping the grayed-out portions just barely discriminable in order to preserve the overall context). Or, you could use color to define the different perceptual groups, each of which should contain no more than four entities.

2. Effort is required to search for information or mentally transform information. Abraham Lincoln was not being complimentary when he said of an acquaintance, "He can compress the most words into the smallest idea of any man I know." An audience should not have to search through a visual or conceptual haystack to find the needle you are talking about.

Example: I once gave a talk in which I argued that it's worthwhile to study a very difficult, perhaps ultimately intractable, scientific problem (understanding the nature of human motivation). In this talk, I used a variant of Pascal's Wager, which is an argument for believing in God. Pascal essentially set up a 2×2 table: The two columns were labeled "God Doesn't Exist" and "God Exists," and the rows were labeled "You Believe in God" and "You Don't Believe in God." I described these columns and rows aloud, and then worked through the four cells, pointing out the crucial cell, which combines "You Don't Believe" and "God Exists." So, the argument goes, given the major downside of being in this cell, and the very small downsides (if any) of the other three cells in the table, you may as well believe. I would then apply this argument to studying my Hard Problem, saying that if it's possible to understand it, we'll lose big-time if we don't try.

However, it turned out that the real "Hard Problem" for the audience was mentally transforming my words into a visual mental image of the 2×2 table, keeping the labels straight in their heads, and then mentally scanning from cell to cell. I often lost the audience, and failed to convert them to my point of view. I solved this problem by actually showing a table (with abbreviated labels), and filling in the cells with plus or minus signs, as appropriate, one by one—ending dramatically with that large minus sign in the last cell (see Figure 1.1). PowerPoint presentations can help people understand by making both memory and processing easier for them.

Presentations that Are Clear and to the Point

Using the eight principles to accomplish the three essential goals—connecting with the audience, directing and holding attention, and promoting understanding and memory—is the foundation for a clear and compelling presentation. Each of these eight principles draws on facts about our mental processes. And each of these principles comes into play, singly or in combination, as we create different aspects of presentations in the following chapters.

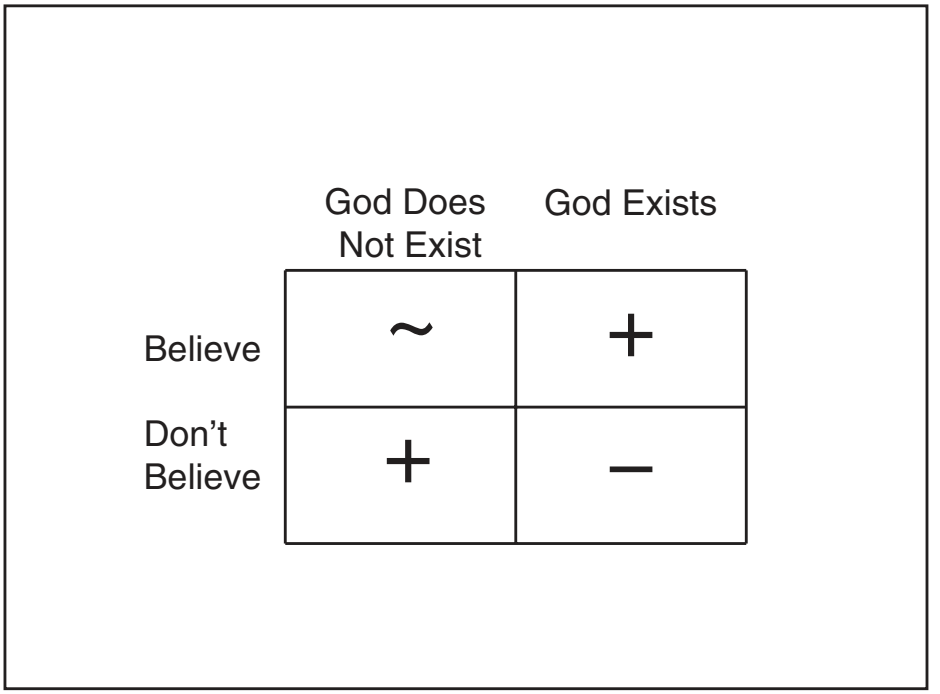

Figure 1.1. A summary of Pascal's Wager, which allows the audience easily to understand and retain the structure of the argument for believing in God.

In the remainder of the book, I will offer many specific recommendations based on these eight principles, some sweeping and some that may seem nit-picking—but all will help you to design and deliver compelling presentations. The principles apply at every level of scale, from the presentation as a whole all the way down to individual illustrations on a single slide and the design of a single bullet point.

Making Points in a PowerPoint Presentation: An Example

The hazards of not respecting the limitations of human mental processes are nicely illustrated—ironically—by one of Tufte's favorite graphics (Figure 1.2). This inventive graphic, created in 1861 by the French engineer Charles Joseph Minard, illustrates Napoleon's bold march to Moscow in 1812 and his dispirited retreat.

This display has never captivated me for the simple reason that—given human processing limitations—I needed several minutes to figure it out. I didn't realize initially that five different variables are being shown simultaneously (which violates the Principle of Capacity Limitations). For one, the width of the line represents the number of soldiers. And I didn't realize that time flows in both directions, left-to-right for the march to Moscow and vice versa for the march back (the lower line). (In our

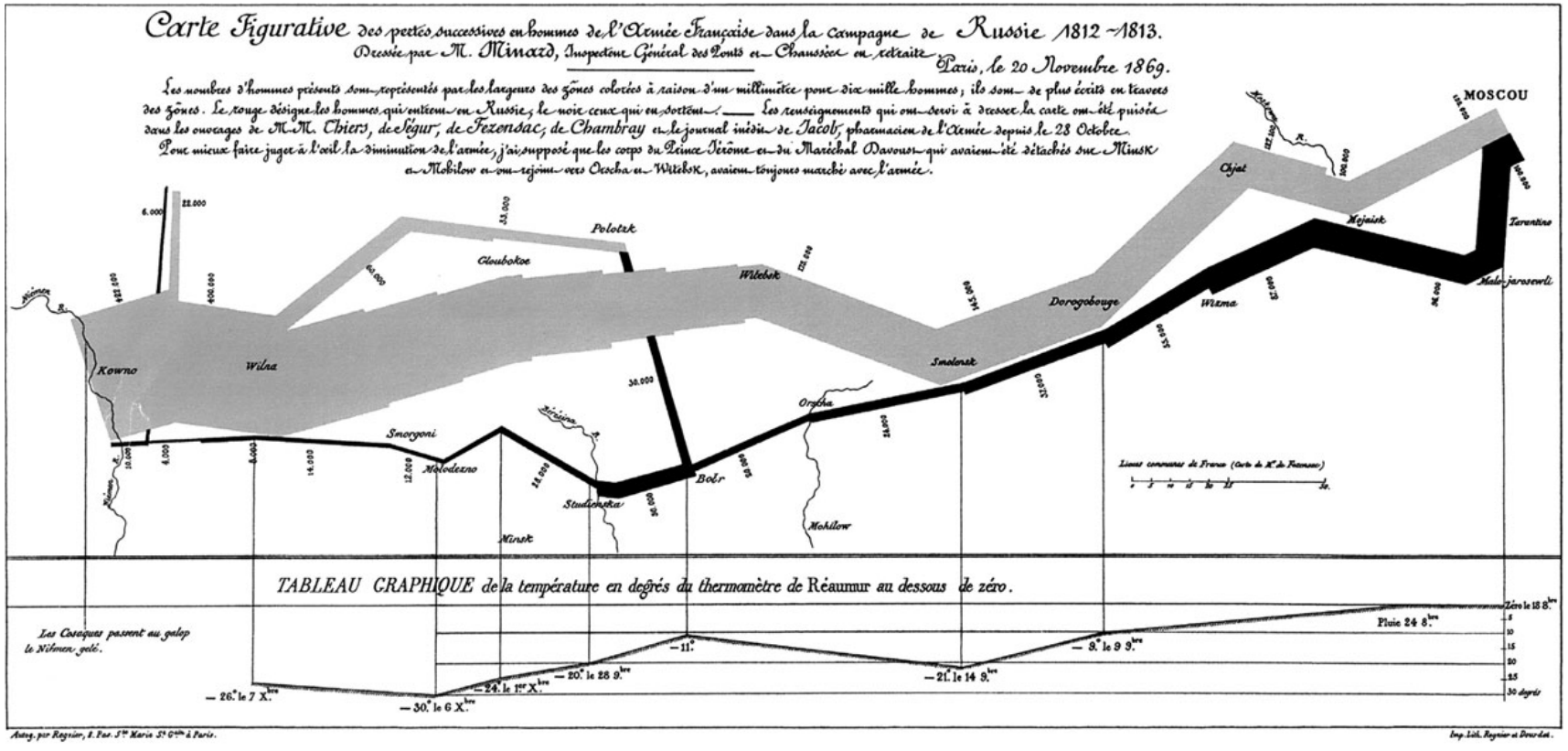

Figure 1.2. Minard's depiction of Napoleon's 1812 Russian campaign (from Tufte, 1983). See also color version, in insert.

culture, time is usually illustrated left to right, and thus the reverse violates the Principle of Compatibility.) Moreover, it took me a while to realize that the line indicating the retreat is indexed by a temperature scale, and I had trouble reading the small numbers that are at odd angles (which presumably indicate the numbers of surviving soldiers; both problems reflect violations of the Principle of Capacity Limitations).

I agree that M. Minard was amazingly clever and managed to cram a huge amount of information into a single display, but I can't agree that this is an effective way to communicate; the display doesn't present the facts so that they're clear or easily absorbed. If you are in the mood, you may enjoy taking the time to study the display for the fun of solving a puzzle, pondering intricate details, or appreciating the graphic devices employed. But if you want the facts and want them in a clear, easily understood way, this display is not the solution.

If you use the eight principles to take advantage of the features of the PowerPoint program, you could create a graphic that would save the viewer a lot of time and trouble: For example, first you could show just the line that represents the march to Moscow (and thereby exploit the Principles of Relevance and Salience). To make obvious the idea of movement over time, you could use animation to extend the line (relying on the Principle of Compatibility). Instead of a solid line, you could use dots—perhaps one for each 1,000 men, arranged into rows (making use of the Principles of Perceptual Organization), marching onward. On the next slide, you could gray out this material, so that it becomes the background for another line, fewer dots wide, moving back to France (thereby exploiting both the Principles of Salience and Discriminability). Indeed, this line could move more slowly (using a sequence of slides to produce variations in the speed of the animation, perhaps using the "custom animation" feature of the PowerPoint program for each horizontal segment), if the return march took more time—which would provide more information than the original, static display. In short, the PowerPoint program would let you direct the viewers' attention to one aspect of the display at a time, emphasize what's important by making the line more distinct than the background, use motion to capture the viewers' attention, and add information—and thus this presentation would be far more effective than the original.

How to Use This Book

Read the chapters in order to open your eyes to a whole new way of making and delivering PowerPoint presentations. Then read the appendix for enough information about the principles to apply them to your every situation.

What's in this Book

The following overview will help you organize what you read in the following chapters.

Chapter 2: Presentations that Work

The next chapter presents a sample PowerPoint presentation (that happens to show how the eight psychological principles can be used to make good graphs). You'll see how to apply the principles to the presentation as a whole: how to organize a presentation to connect with the audience, direct and hold attention, and promote understanding and memory.

Chapter 3: Legible Text

We next turn to the nitty-gritty of how to prepare individual slides. Here, you'll get guidelines for the appropriate sizes and fonts of text, and the use of bulleted lists. You'll also find out when to use tables and how to construct them so that they convey information effectively.

Chapter 4: Color, Texture, Animation, and Sound

Color, texture, animation (i.e., movement on the screen), and sound can bring a presentation to life and emphasize or enhance specific aspects of your message. Used incorrectly, however, they can distract and can bury specific aspects of your message. You'll see here how to use color to highlight material, convey information, and organize the display effectively. You'll also learn which colors tend to work well together and which don't, and what emotional associations specific colors convey. Then you'll learn how to use texture variations (in figure and background) and animation. And you'll learn the pro's and con's of adding specific types of soundtracks to your presentations.

Chapter 5: Communicating Quantitative Information: Graphs

Graphs are pictures that convey information about quantities. Because well-constructed graphs can communicate a lot of information quickly and effectively (perhaps even being worth the proverbial 1,000 words), they often are used in presentations. You'll see when to use graphs and which sort to use to illustrate a given type of data, to make a specific point, and to connect with a specific audience. You'll also get guidelines for how to make different sorts of graphs.

Chapter 6: Communicating Qualitative Information: Charts, Diagrams, Maps, Photographs, and Clipart

Some types of graphics—such as organizational charts and process diagrams—do not convey quantitative information. But the same principles that apply to quantitative graphs will also help you create

effective qualitative graphics. You'll also see how to choose an appropriate icon or illustration, and when such visual aids will drive a point home or just drive the viewer to distraction.

Chapter 7: The Good, the Bad, and the Incomprehensible
In this final chapter, I begin by taking a single display that was once a candidate to be on every food item in the United States and analyze it using the eight principles. This analysis illustrates the power of the recommendations reviewed in the previous pages. Then we'll look at problems with certain features of the PowerPoint program, including cases when defaults either aren't helpful or are downright destructive. We'll finish by coming back to Tufte's attack on the PowerPoint program. I defend the potential of the medium and urge the reader not to blame the messenger medium for bad messages.

Appendix: Psychological Principles and Their Specific Aspects

Here you'll get a more detailed treatment of the principles, which provides you with enough information to apply them to all aspects of presentations.

From the Principles to the Point

At the end of each chapter I summarize the major points by bringing them back to the principles. This allows me to avoid citing the individual principles repeatedly in the body of the text and also reinforces the crucial role the principles play in these recommendations.

In this chapter, you saw that an effective presentation accomplishes three goals: It connects with the audience, directs and holds attention, and promotes understanding and memory. You also learned that this book presents a set of specific recommendations for achieving these goals; these recommendations are grounded in principles about the operation of the mental processes that underlie perception, memory, comprehension, cognition, emotion, and motivation. Each subsequent goal draws on the previous principles as well as any additional ones.

My advice focuses on how to structure your presentation so that your audience will receive and retain your message. Although there are no formulae that guarantee that you will convince your audience,

I can tell you this: If you ignore the three goals and the eight principles, you may as well be getting ready to run a foot race with one leg tied behind your back.

The three goals and principles that address them are as follows; each goal draws on the previously mentioned principles as well as the ones added specifically to address that goal.

Goal 1: Connect with your audience

Principle 1: The Principle of Relevance
Communication is most effective when neither too much nor too little information is presented.

Principle 2: The Principle of Appropriate Knowledge
Communication requires prior knowledge of relevant concepts, jargon, and symbols.

Goal 2: Direct and hold attention

Principle 3: The Principle of Salience
Attention is drawn to large perceptible differences.

Principle 4: The Principle of Discriminability
Two properties must differ by a large enough proportion or they will not be distinguished.

Principle 5: The Principle of Perceptual Organization
People automatically group elements into units, which they then attend to and remember.

Goal 3: Promote understanding and memory

Principle 6: The Principle of Compatibility
A message is easiest to understand if its form is compatible with its meaning.

Principle 7: The Principle of Informative Changes
People expect changes in properties to carry information.

Principle 8: The Principle of Capacity Limitations
People have a limited capacity to retain and to process information, and so will not understand a message if too much information must be retained or processed.